Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

Lots of if statements, the user input function, drafter, (i would like to improve my drafter skills because I think that and heavily nested data are my weakest areas), probably dataclasses to organize the different options,

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

I think the easiest part of the project will be the after start option of seeing how the wastewater treatment part works, because Im keeping it general and just going step by step of the overview of the process. I think the hardest part will be if I do decide to implement a groundwater testing part of the program, that will have a couple different outcomes that might be hard to keep track of.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

I think drafter is going to take a large portion of time, so Im hoping to get all the functions and whatnot out of the way so I can spend lots of time on that and allow myself time to get help if need be

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

Some distractions include myself and thinking about other class work I have, and Im gonna minimize that by putting the cisc tab/group I have open on my computer on an entirely different window thing from all my other work and remove my to do list stickynote from my computer so I can focus on this for the allotted amount of time I decided. I am using online resources like the EPA and the CDC websites to fact check my information about the different cycles and processes involved in my project, and I noticed that the first assignment for this said something about the website solving a problem, but my website is just going to inform people about how different kinds of water get treated. Is that okay?

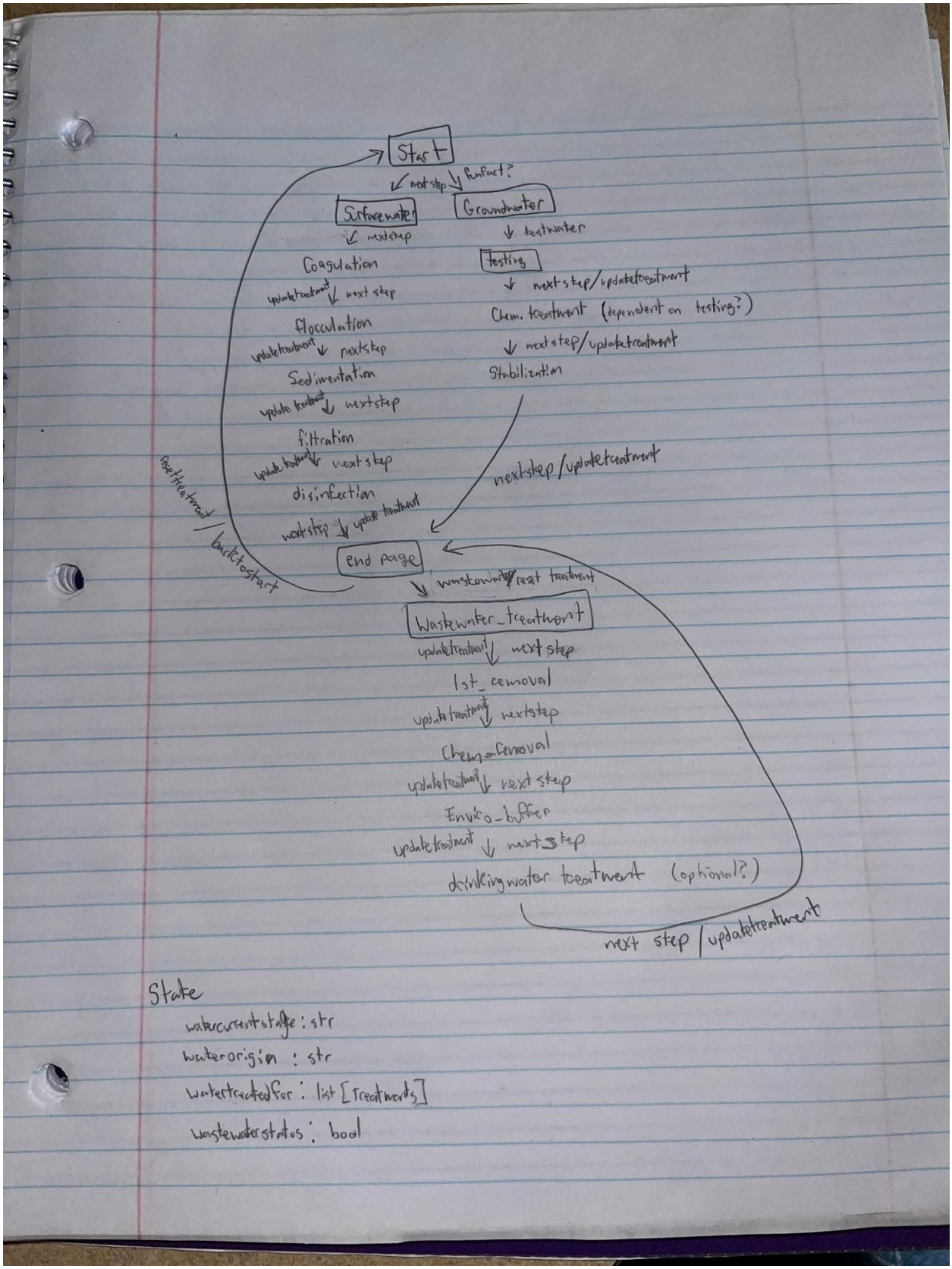
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.



State

watercurrentstate : str

waterorigin : str

Watertreatedfor : list [treatments]

Wastewaterstatus : bool

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	The purpose of my site is to inform people about something they may not think about that they use everyday, water, and tell them about the processes it goes through when treated. It will have 2 start options (i think), either ground water or surface water, and from there it will walk you through the interactive process of treatment, and allow you to see what happens after the water is used if you would like once you treated the water and send it off to households to use.
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	watercurrentstage: str Updates with current stage the water being treated is in waterorigin: str Identifies where the water came from (?) what type of water the user started with watertreatedfor: list[Treatments] A list of what the water has already been treated for wastewaterstatus: bool If the water is wastewater or not (changes based on if wastewater treatment page is opened)
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	Treatments - multi/chem, description, orderinprocess, name
Routes (minimum 7)	Nextstep - takes user to the next step in water

The names of your route functions	treatment process (groundwater and surface water?) Funfact - displays facts about different steps of water treatment process, relevant to what process step user is in Testwater - takes user to page where groundwater gets tested for different chemicals (?) Updatetreatments - updates list of treatments displayed at page marking where in treatment process user is (?) Backtostart- takes user back to start from endpage if they dont want to see how wastewater is treated if they want to pick a different path through water treatment process Wastewater - takes user from endpage to wastewater process walkthrough Resettreatment - resets treatment list displayed on page marking where in treatment process user is (?)
Pages (minimum 5) A brief description of the pages in your site	Startpage - displays options for what path user would like to start on, and maybe brief description of what this does/introduction to water treatment process Groundwater - page option after start page, gets updated with different treatment process steps as it goes through groundwater treatment process Testing - part of groundwater treatment process, tests for the 3 different things and displays results Surfacewater - page option after start page, gets updated with different treatment process steps as it goes through surface water treatment process Endpage - displays options user can do after walking through any type of water treatment process (backtostart, wastewater) Wastewatertreatment - page option after endpage, gets updated with different treatment process steps as it goes through wastewater treatment process
User Input (minimum 3)	Whattesting - textbox, asks what kind of testing (3 options displayed on testing page) the user would like to do first

Description of user input components (TextBox, TextArea, CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	Wheretonext - selectbox that allows users to choose (from the endpage) where they go next in the website Funfact? - textbox, asks user if they would like a fun fact about the current step in the water treatment process (answers are yes or no)
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	For treating page if statement will need to be used to determine result of what kind of test the user gets (3 options, if one test done display this result, if a different one done display different result) For determining where the user goes after endpage - if wastewater is selected they go to the wastewater page, if backtostart is selected they go back to start For determining how the watercurrentstage and watertreatedfor attributes of the state dataclass gets updated (if nextstep is coagulation for surfacewater page, then watercurrentstage = coagulation, and added to watertreatedfor list)
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	watertreatedfor list gets looped through by resettreatment route/function and saved in a previouswalkthrough instance of state dataclass, reserving past walkthroughs of treatment process before resetting watertreatedfor list for next walkthrough

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

No llm was used for this assignment

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.